

Lakshay Kumar

Machine Learning Engineer | LLM Systems | Multimodal AI | C++ Simulation

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PROFILE SUMMARY

Machine Learning Engineer with internship-backed work across LLM serving, RAG systems, multimodal retrieval, OCR benchmarking, FastAPI AI services, and C++/VTK biomechanics simulation. Known for turning ambiguous research or prototype work into measurable engineering evidence: vLLM load tests, retrieval benchmarks, mesh-quality fixes, multilingual assistant paths, and headless documentation automation.

EXPERIENCE

Machine Learning Intern | CreateBytes

Sep 2025 - Present

Remote / India

- Migrated and stabilized a Chainlit legal assistant on vLLM with prefix caching, 5-turn memory, streaming synchronization fixes, and load testing up to 50 users.
- Benchmarked multimodal retrieval, captioning, OCR, and FAISS index strategies across 1,000 video-caption pairs, 104 manually annotated CCTV videos, and 600k embeddings.
- Implemented FastAPI document-assistant paths for PDF/JSON/text parsing, multilingual summarization, context summarization, and CPU audio processing; resolved OOM failures during profiling.
- Delivered C++/VTK/QML FE-HBM repositioning improvements for Piper, including metadata corrections, contour fixes, negative-element diagnostics, and headless report/image automation.

TECHNICAL SKILLS

AI / ML: RAG, LLM fine-tuning, vLLM, Prefix caching, LangChain, ChromaDB, QLoRA, LoRA, BERTScore

Programming: Python, C++, Java, JavaScript, TypeScript

Simulation / Tools: VTK, Qt/QML, CMake, LS-DYNA, CGAL, SWIG, Linux, Git/GitHub, Postman

Multimodal / CV: InternVideo2, LanguageBind, X-CLIP, Gemini Embedding 2, PaddleOCR-VL, OpenCV, YOLO, MediaPipe, FAISS

Backend / Web: FastAPI, React, Node.js, Express, REST APIs, Streamlit, MongoDB, MySQL

Core: Data Structures, Algorithms, OOP, Operating Systems, Computer Networks, DBMS

PROJECTS

Legal Assistant Optimization on vLLM | vLLM, Chainlit, RAG, Prompt Engineering, Load Testing

- Migrated inference to vLLM, enabled prefix caching, and fixed EOS latency plus background-stream synchronization issues.
- Validated parallel behavior up to 50 users and observed >36 tok/s in 10-user tests while improving memory handling and edge-case responses.

Multimodal Retrieval and OCR Benchmark Suite | InternVideo2, LanguageBind, X-CLIP, Gemini Embedding 2, PaddleOCR-VL, FAISS

- Benchmarked model families across 1,000 video-caption pairs and 104 annotated CCTV videos using recall, confidence, and caption-quality signals.
- Compared FAISS Flat/HNSW/IVF indexes at 600k embeddings and evaluated OCR stacks for scanned PDFs and vehicle/video datasets.

STORMS Document and Multilingual Assistant | FastAPI, Qwen, Gemma, Indic2 200M, vLLM, Audio Processing

- Built FastAPI service paths for PDF/JSON/text parsing, summarization, chatbot flows, multilingual support, and context summarization.
- Integrated Indic2 200M for five-language translation support and resolved CPU audio OOM failures during performance profiling.

Piper FE-HBM Computational Biomechanics | C++, VTK, Qt/QML, CMake, LS-DYNA, CGAL, SWIG

- Corrected metadata, contour, and timestep issues to extend stable knee flexion to 70 degrees and reduce GHBM negative elements from roughly 6000 to 2.
- Built negative-volume visualization and a C++/Python headless documentation pipeline for repeatable FE-HBM evidence capture.

EDUCATION

Chitkara University

B.E. in Computer Science (CGPA 8.87)

Oct 2022 - Jun 2026
Patiala, Punjab

GMSSS, Mani Majra

Class 12 CBSE (95.4%)

2020 - 2021
Chandigarh

Blue Bird High School

Class 10 CBSE (95.2%)

2018 - 2019
Panchkula, Haryana

ACHIEVEMENTS

- Class Representative at Chitkara University, coordinating communication and team activities across the class.
- Consistent DSA practice across LeetCode, GeeksforGeeks, W3Schools, and Coding Ninjas.
- Comfortable working across research-heavy ambiguity, product UI/API delivery, and low-level C++ debugging.